



**Programme: B E – Computer Science and Engineering (AI&ML)
Computer Science and Engineering (Cyber Security)
Internal Assessment – I**

TERM : NOV to MARCH	COURSE NAME : Database Management System
DATE : 03-01-2024 TIME: 3.30-4.30	COURSE CODE : CY32/CI32
MAX MARKS: 30	PORTIONS : 1 st & 2 nd Module



Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.**

Marks: 15x2=30

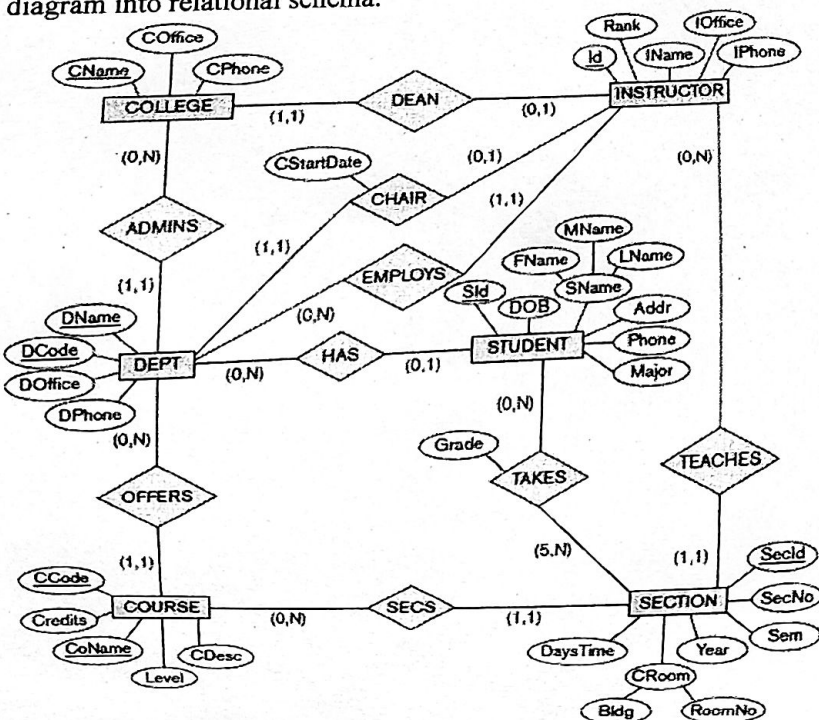
Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Explain the characteristics of database systems.	L2	CO1	4
1.b	Classify any three Actors on the scene. Explain each with their responsibilities.	L1	CO1	4
1.c	With neat diagram explain three schema architecture and need of mapping between schema levels.	L1	CO1	7
2.a	Explain the Following 1. Super Key 2. Primary Key 3. Candidate Key 4. Foreign Key.	L2	CO2	3
2.b	Draw ER Diagram for an employee database. The scenario are as follows. 1. An employee works for a department. 2. Every department is headed by a manager. 3. An employee works on one or more projects. 4. An employee has dependents. 5. A department controls the project.	L3	CO1	5
2.c	Demonstrate mapping of binary 1:N relationship and binary M:N relationship types of ER diagrams to relations. Convert the figure 2(c) ER diagram into relational schema. 	L2	CO2	7

Figure 2(c): ER Diagram of College-Student



3.a	Demonstrate the update operations, transaction and dealing with constraints violations with suitable examples.	L2	CO2	5
3.b	Differentiate between entity integrity and referential integrity constraints. Why are they important?	L2	CO2	3
3.c	Consider the schema given below: Student (USN, Name, Branch, CGPA) Faculty (F_ID, Name, Department, Designation, salary) Course (C_ID, Name, F_ID) Enroll (C_ID, USN, Grade) Give the relation algebra expression for the following: i. List the USN and names of students who have secured 'B' grade in CS513 course. ii. List the names and salaries of faculties who are professors. iii. For each department, find the maximum salary of instructors in that department. iv. List the courses and the number of students enrolled for that course.	L3	CO2	7

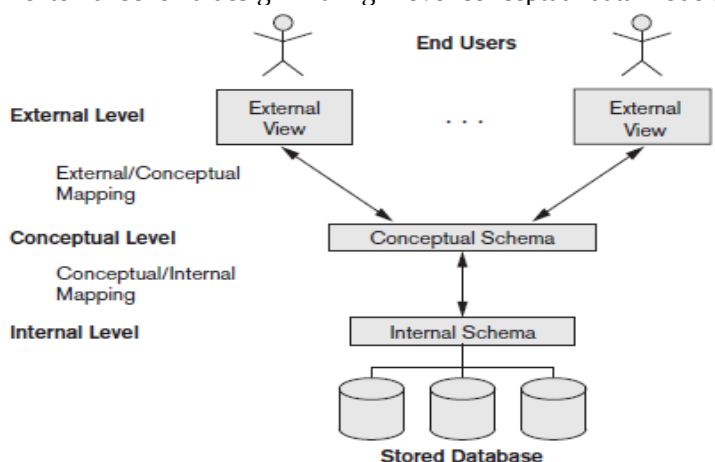
* **L1** – Remember, **L2** – Understand, **L3**- Apply, **L4**- Analyze, **L5**-Evaluate, **L6**-Create

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Answer Scheme
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Q. NO	Answers	Marks
1.a	In the database approach, a single repository maintains data that is defined once and then accessed by various users repeatedly through queries, transactions, and application programs. The main characteristics of the database approach versus the file-processing approach are the following: ■ Self-describing nature of a database system A fundamental characteristic of the database approach is that the database system contains not only the database itself but also a complete definition or description of the database structure and constraints. ■ Insulation between programs and data, and data abstraction In traditional file processing, the structure of data files is embedded in the application programs, so any changes to the structure of a file may require changing all programs that access that file. By contrast, DBMS access programs do not require such changes in most cases. The structure of data files is stored in the DBMS catalog separately from the access programs. We call this property program-data independence. ■ Support of multiple views of the data A database typically has many types of users, each of whom may require a different perspective or view of the database. A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored. Some users may not need to be aware of whether the data they refer to is stored or derived. ■ Sharing of data and multiuser transaction processing A multiuser DBMS, as its name implies, must allow multiple users to access the database at the same time. This is essential if data for multiple applications is to be integrated and maintained in a single database. The DBMS must include concurrency control software to ensure that several users trying to update the same data do so in a controlled manner so that the result of the updates is correct. For example, when several reservation agents try to assign a seat on an airline flight, the DBMS should ensure that each seat can be accessed by only one agent at a time for assignment to a passenger.	1 1 1 1
1.b	Actors on the Scene For a small personal database, such as the list of addresses discussed in previously one person typically defines, constructs, and manipulates the database, and there is no sharing. However, in large organizations, many people are involved in the design, use, and maintenance of a large database with hundreds or thousands of users. Database Administrators In any organization where many people use the same resources, there is a need for a chief administrator to oversee and manage these resources. In a database environment, the primary resource is the database itself, and the secondary resource is the DBMS and related software. Database Designers Database designers are responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store this data. These tasks are mostly undertaken before the database is actually implemented and populated with data. End Users End users are the people whose jobs require access to the database for querying, updating, and generating reports; the database primarily exists for their use. There are several categories of end users:	1 1 1

	Casual end users occasionally access the database, but they may need different information each time. Naive or parametric end users make up a sizable portion of database end users. Sophisticated end users include engineers, scientists, business analysts, and others who thoroughly familiarize themselves with the facilities of the DBMS in order to implement their own applications to meet their complex requirements. ■ Standalone users maintain personal databases by using ready-made program packages that provide easy-to-use menu-based or graphics-based interfaces. An example is the user of a financial software package that stores a variety of personal financial data.	1
1.c	The Three-Schema Architecture The goal of the three-schema architecture, illustrated in Figure, is to separate the user applications from the physical database. In this architecture, schemas can be defined at the following three levels: 1. The internal level has an internal schema, which describes the physical storage structure of the database. The internal schema uses a physical data model and describes the complete details of data storage and access paths for the database. 2. The conceptual level has a conceptual schema, which describes the structure of the whole database for a community of users. The conceptual schema hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints. 3. The external or view level includes a number of external schemas or user views. Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group. As in the previous level, each external schema is typically implemented using a representational data model, possibly based on an external schema design in a high-level conceptual data model. 	1 1 2 3
2.a	1. Super Key: A super key is a set of one or more attributes (columns) that, taken collectively, uniquely identify a record within a table. It's a superset of a key and may contain additional attributes apart from the minimum required to uniquely identify a record. 2. Primary Key: A primary key is a specific type of super key that uniquely identifies each record in a table and ensures there are no duplicate records. It must contain unique values and cannot have NULL values. Each table can have only one primary key. 3. Candidate Key: A candidate key is a set of attributes that can uniquely identify a tuple/record in a table. It meets the criteria of uniqueness and irreducibility—meaning no subset of these attributes can uniquely identify a record. From these candidate keys, one is chosen as the primary key. 4. Foreign Key: A foreign key is a field in one table that references the primary key in another table. It establishes a link between two tables by enforcing referential integrity. This means the values in the foreign key must match values in the primary key column of the referenced table or be NULL. In essence, primary keys uniquely identify records within a table, foreign keys establish relationships between tables by referencing the primary key of another table, super keys are sets	1 1 1



	of attributes that uniquely identify records (which may include more attributes than needed), and candidate keys are potential choices for the primary key within a table.	
2.b		5
2.c	Mapping of binary 1:N relationship and binary M:N relationship types of ER diagrams to relations are as follows. Mapping binary 1:N relationships: 1. Create a separate relation for each entity type. 2. In the relation for the entity type on the "N" side of the relationship, include the primary key of the entity type on the "1" side as a foreign key. Mapping binary M:N relationships: 1. Create a separate relation for each entity type. 2. Create a new relation to represent the relationship itself. 3. This new relation will have two foreign keys, one referencing the primary key of each entity type involved in the relationship. 4. The combination of these foreign keys will form the primary key of the new relation. figure 2(c) ER diagram into relational schema. • COLLEGE (<u>CName</u>, CPhone, COffice) • CHAIR (<u>Id</u>, CStartDate) • DEPT (<u>DName</u>, <u>DCode</u>, DOffice, DPhone) • INSTRUCTOR (<u>Id</u>, IName, IPhone, Rank, IOffice) • STUDENT (<u>SI</u>, FName, MName, LName, Addr, Phone, DOB, Major, <u>DCode</u>, Grade) • TAKES(Grade) • COURSE (<u>CCode</u>, Credits, <u>CoName</u>, Level, CDesc) • SECTION (<u>SecID</u>, SecNo, Sem, Year, Bldg, RoomNo, DaysTime)	1 2 4

3.a	The Update (or Modify) operation is used to change the values of one or more attributes in a tuple (or tuples) of some relation <i>R</i>. It is necessary to specify a condition on the attributes of the relation to select the tuple (or tuples) to be modified. Here are some examples. ■ Operation: Update the salary of the EMPLOYEE tuple with Ssn = '999887777' to 28000. <i>Result:</i> Acceptable. ■ Operation: Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 1. <i>Result:</i> Acceptable. ■ Operation: Update the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 7. <i>Result:</i> Unacceptable, because it violates referential integrity. ■ Operation: Update the Ssn of the EMPLOYEE tuple with Ssn = '999887777' to '987654321'. <i>Result:</i> Unacceptable, because it violates primary key constraint by repeating a value that already exists as a primary key in another tuple; it violates referential integrity constraints because there are other relations that refer to the existing value of Ssn. Updating an attribute that is <i>neither part of a primary key nor part of a foreign key</i> usually causes no problems; the DBMS need only check to confirm that the new value is of the correct data type and domain. Modifying a primary key value is similar to deleting one tuple and inserting another in its place because we use the primary key to identify tuples.	5
3.b	1. Entity Integrity Constraint: This constraint ensures that each row or record in a table is uniquely identified and that the primary key column(s) do not contain any NULL values. In simpler terms, it guarantees the uniqueness of the primary key within a table and prevents it from having null values. This constraint is maintained by the definition of the primary key. 2. Referential Integrity Constraint: Referential integrity ensures the consistency of data between tables. It is maintained through the use of foreign keys. It dictates that values appearing in a foreign key column of one table must exist in the primary key column of another table (or be NULL, depending on the setup of the foreign key column). Importance: • Data Accuracy: These constraints ensure accuracy and consistency within the database. Entity integrity prevents duplicate records and ensures each record is uniquely identifiable. Referential integrity maintains the relationships between tables, preventing orphans (records with references to non-existing records). • Data Reliability: By enforcing these constraints, databases become more reliable. They reduce the chances of data corruption, inconsistencies, and logical errors that could arise due to incomplete or inaccurate relationships between tables. • Data Integrity: Integrity constraints safeguard the integrity of the database by defining rules that must be followed during data manipulation operations like insert, update, or delete. This prevents invalid or incorrect data from entering the database. Overall, these constraints play a crucial role in maintaining the quality, accuracy, and reliability of data within a database, ensuring its usability and trustworthiness for various applications and operations.	1 1 1

3.c	i. List the USN and names of students who have secured 'B' grade in CS513 course. (γ – Aggregate and Grouping function.) $R1 = \pi \text{ USN, Grade } (\sigma \text{ Grade} = 'B' \text{ and } C_ID = 'CS513' (\text{Enroll}))$ $(\text{Enroll} \bowtie \text{Student})$ $R2 = \pi \text{ USN, Name } (\text{Enroll. USN} = \text{Student. USN})$	2
	ii. List the names and salaries of faculties who are professors. $R1 = \pi \text{ Name, salary } (\sigma \text{ Designation} = 'Professor' (\text{Faculty}))$	1
	iii. For each department, find the maximum salary of instructors in that department. $R1 = \pi \text{ Department, Salary } (\sigma \text{ Designation} = 'Instructor' (\text{Faculty}))$ $R2 = \text{Department } \gamma \text{ MAX (salary)}(R1)$	2
	iv. List the courses and the number of students enrolled for that course. $R1 = C_ID \gamma \text{ COUNT(USN)} (\text{Enroll})$	2